

Li-Fi technology in Traffic light

V.K.G.Kalaiselvi^{#1}, A.Sangavi^{#2}, Dhivya^{#3}

^{#1}Assistant Professor, Information Technology

^{#2}Student, Department of Information Technology

^{#3}Student, Department of Computer Science

^{#1#2}SriSairamEngineeringCollege

^{#1}vkghalaiselvi@gmail.com ^{#2}sangaviaravindan128@gmail.com

^{#3}dhivi.admu@gmail.com

Abstract

Wireless communication has become a basic utility in personal and business life such that it becomes a fundamental of our lives and this communication uses the radio spectrum for data transfer. There are issues in using the radio spectrum further, they are, capacity, efficiency, availability and security. The defects of Wi-Fi technology has given birth to the concept of **Li-Fi (Light Fidelity)** technology. Li-Fi can be defined as a light based Wi-Fi. This technology mainly serves the purpose of transmitting data using retrofitting of LED Bulbs that has high efficiency, durability and reliability. This technology emerged by 1990s in a lot of countries like Japan, German and Korea. The proposed paper aims in using the Wi-Fi technology and enabling communication of vehicles with the traffic light system in order to prioritize the vehicles and change the signals accordingly rather than by a process of pre-defined order or by manual order. Traffic lights already use LED lighting, so that this proposed system may seem easy to implement. Sending data through siren lights in an ambulance and fire extinguishers to a traffic light control system and switching the signal in order to allow faster and non-interrupted transport. Here is already a kind of system in play for GPS navigational systems, but the traffic lights can be used in updating drivers using basic information or streaming video directly from news broadcasts i.e, about the accidents or delays up ahead.

Keywords: Li-Fi , traffic light management, LEDs

1.INTRODUCTION

There are around 1.4 million cellular mast radio waves base stations set, with over 5 billion mobile

phones. Mobile phones transmit over 600TB of data on an average basis for every month. Nowadays wireless communication use radio waves. But radio waves have a problem of efficiency, availability, security and capacity. Spectrum is very important requirement for wireless communication. With advancement in technology and increase in number of users, existing radio wave spectrum fails to meet the need and hence, the capacity problem. To resolve all the issues, we have come up with the concept of transmitting data wirelessly through light using LEDs, called as Li-Fi which is a latest technology that makes use of LED lights which helps in the transmission of data much more faster, and flexible because of the durability, efficiency and high lifetime characteristics that makes Li-Fi concept a better one. LED lights are nowadays widely used for personal and official purposes for their luminous efficacy improvement. Visible light communication (VLC) is a new way of wireless communication using visible light. Typical transmitters used for visible light communication are visible light LEDs and receivers are photodiodes and image sensors. Being a highly populated country like India and lot of traffic problems, there is always a problem of manual traffic control whenever an ambulance arrives along a particular direction which is not effective. Automatic traffic control and uninterrupted traffic for fire extinguishers and ambulance is highly necessary. The proposed technology Li-Fi by which the transmission of data can be enabled between the lights of traffic light system and that of the lights in the siren of ambulance and fire extinguishers on a priority-basis will enable us to have a better traffic control system with un-interrupted services of transport for the ambulances.

2) DESIGN OF Li-Fi

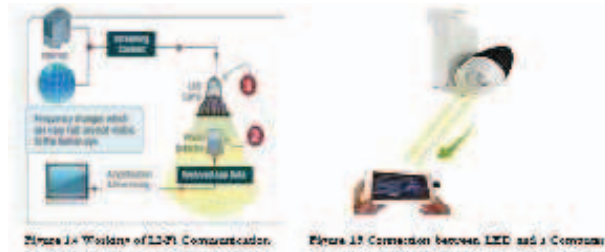
Li-Fi architecture will require many LED bulbs ,wireless devices such as PDA, Mobile Phones, and laptops. Important factors we should consider while designing Li-Fi as following:

- Presence of Light
- Line Of Sight

- Devices: fluorescent light & LED

The streaming content should have proper integration with server and internet network to have an efficient working.

3) WORKING OF LI-FI



The first diagram describes how the binary data is getting captured by the light receptors that are installed in all types of connected devices for example phones, tablets, desktops to any appliance. Matter experts state clearly that light pulses are imperceptible to human eyes, without any damage. In addition, any lamp or flashlight can become a hotspot. By having an LED in one hand and a detector in other hand, we can understand the working of Li-Fi. If the LED is ON, the photo detector registers a binary 1, else it's a binary 0. Flash the LED enough times and you build up a message of Ones and Zeroes. Usage of different color LEDs will have higher data rates may be leading to megabits per second, this is accomplished by the flickering of LEDs to create binary code (on = 1, off = 0), and is done at higher rates than the human eye can detect. The more LEDs in your lamp, the higher data it can process.

The second diagram shows connection between the information from a computer and internet with LED. One LED transfers data at a considerably slower rate and hence millions of LEDs with one micron size are installed in the bulb. The reduction of size of LEDs increases the efficiency of one light bulb to transmit the data at unexpectedly higher rates. Also, these micro-LEDs are ultimately just pixels and at one micron, these LEDs would be a lot smaller than those in the Smartphone's retina display. You could have a huge array of these LEDs that double up as a room's light source and a display and provides networking capability on the side.

4) ADVANTAGES OF Li-Fi:

- Green information technology: Li-Fi technology has no side effects on any living things unlike radio-waves which give out harmful radiations that affects birds, human beings etc.
- No Frequency Bandwidth Problem: Li-Fi is a visible light communication medium, so need of spectrum license is unnecessary. We don't

have to pay any amount for communication and license issues.

- High Communication Security: Light cannot infiltrate through the walls in Li-Fi technology; hence security between walls is highly good. Transmission of data is blocked through walls is blocked thereby avoiding data leakages.
- Bandwidth of Li-Fi is 10,000 times more than the radio waves.
- Capacity: As there is being increasing number of users and newer 2G, 3G technologies, the spectrum is running out of capacity. And comparatively VLC has a higher spectrum than the present radio waves spectrum.
- Efficiency: There are almost 14 lakhs cellular base stations situated around but the percentage of efficiency from each base station is less than 6% Li-Fi will be efficient as it will consume very less energy.
- Availability: There is no need to switch off mobiles in aircrafts or near petrol pumps. As light is present everywhere, data is too present.

5) CHALLENGES OF Li-Fi

- Li-Fi technology has problems of Line Of Sight problem which means no disturbance in the path of transmission.
 - Also the problem of receiver transmitting back to the transmitter is a big issue.
 - Visible light cannot penetrate through walls (radio waves do) and hence a simple disturbance can block the data that is being transmitted.
 - Reliable property and network coverage are serious problems to be considered because the interference from external light sources like sun light, bulbs and opaque materials in the path of transmission results in disturbance in the communication.
 - Initial installing of the VLCs may seem higher in cost when implementing on a large scale but the further operating costs like the maintenance and electricity charges will get reduced. It is an affordable and reliable project.
 - Li-Fi technology is not possible along highly terrain area or a remote area where there are more probabilities in the obstructions. Thereby affecting the LOS (Line of Sight)

6) APPLICATIONS OF Li-Fi

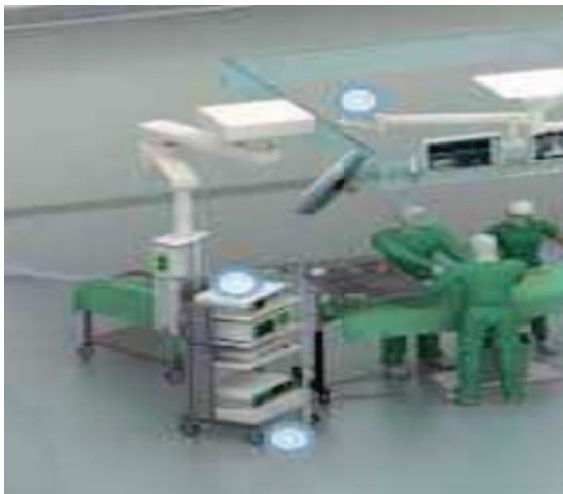
- FIELD OF AIRWAYS TRANSPORT

The problem of communication media at the time of travelling in airplanes because the whole of communication in airways is done on the basis of radio waves, now can be overcome by the usage of Li-Fi as it does not interfere with any radio-equipments.



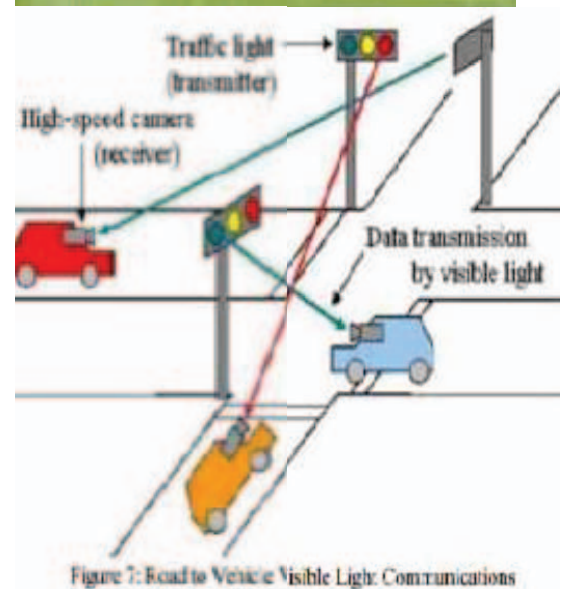
- MEDICAL FIELDS

Operating rooms will not have access to any equipment concerning radiations. Also the fear of interference of signals from handheld device and desktops can block signals from monitoring equipments. The Operating rooms will definitely have access to LEDs, hence implementation of Li-Fi is easy.



- TRAFFIC CONTROL

Traffic lights also can communicate to the car and communication can be established even between 2 cars as cars have front and back LED lights and thereby preventing or at least decreasing accidents.



- UNDERWATER VEHICLES

Remotely Operated underwater Vehicles don't operate well when tether isn't too long to explore an area or when it gets struck onto something. If at all wires are replaced with light they would be much free to explore. Headlamps may also used to communicate, processing data and referring findings back to surface.

